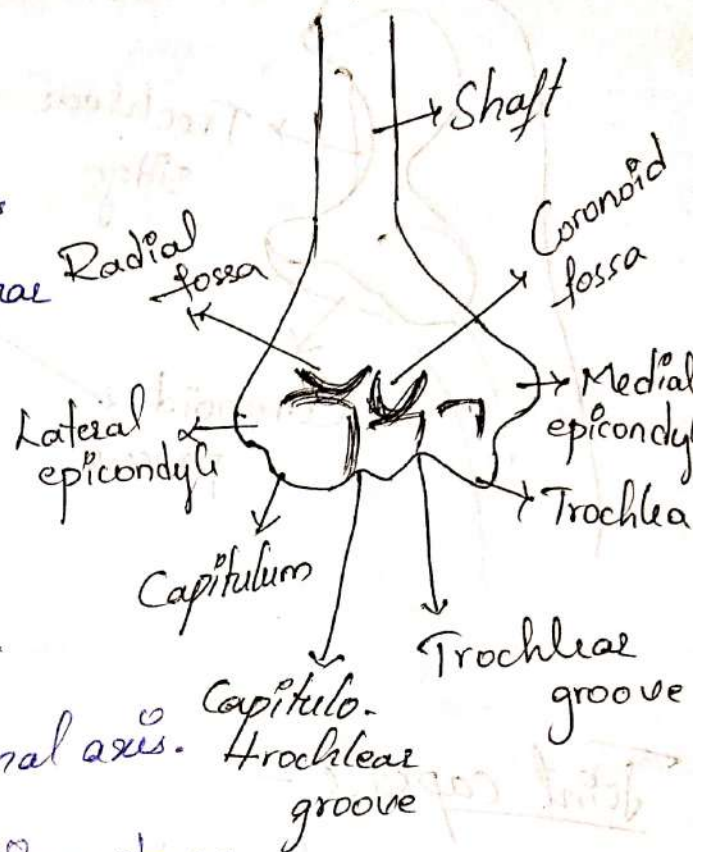


Elbow Complex

Articular Surface :-

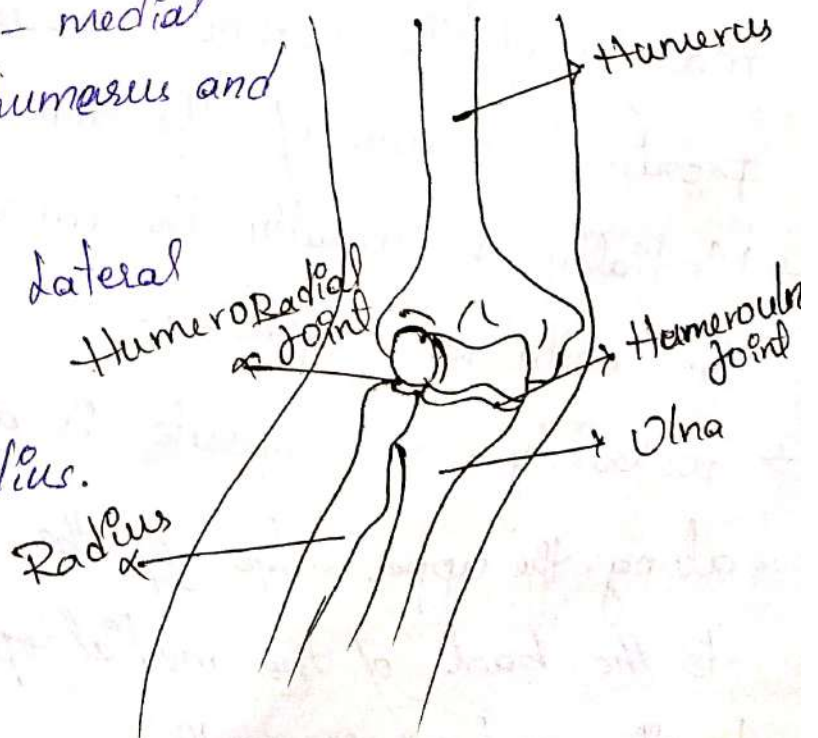
Introduction -

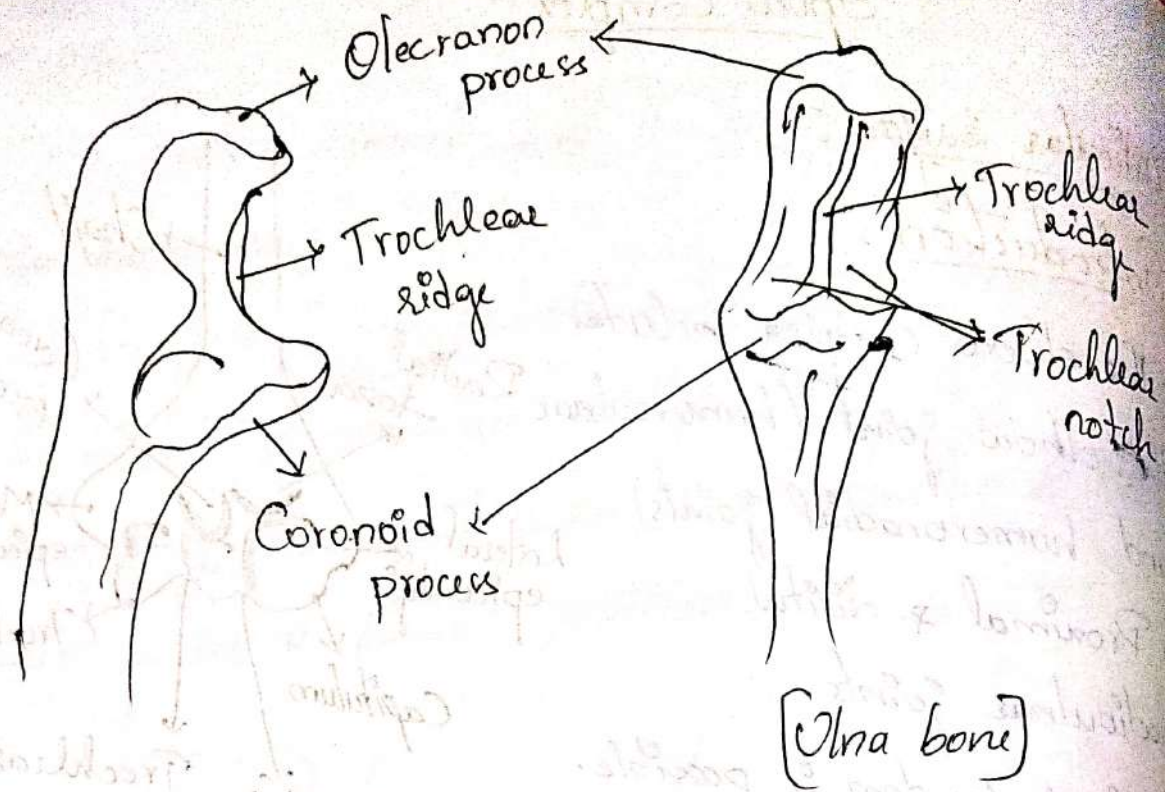
- The elbow complex includes the elbow joint (humeroulnar and humeroradial joints)
- Proximal & distal radioulnar joints.
- 1° of freedom is possible.
- Sagittal plane with coronal axis.
- It is a plane variety of hinge type joint & uniaxial variety.



Articular Surface -

- Proximal - Inferior surface of the humeral condyle.
- Distal - Head of the Radius and Ulna
- Humeroulnar joint - medial aspect of the distal humerus and the ulna.
- Humeroradial joint - lateral aspect of the distal humerus and the radius.





Joint capsule -

- The humeroulnar, humeroradial and superior radio-ulnar joint, are enclosed in a single joint.
- Proximal - Humeral attachment is just below the above the coronoid and radial fossae.
- Distal - Capsule attached into the ulna along the margin of the coronoid process & bends with the proximal border of the annular ligament.
- Medially & laterally the capsule is continuous with the collateral ligaments.
- Posteriorly - the capsule is attached to the humerus along the upper edge of the olecranon fossa and to the back of the medial epicondyle. and attaches to the neck of the radius.

Ligaments -

- 1) Medial (ulna) Collateral Ligament
- 2) Lateral (Radial) Collateral Ligament

1) Medial Collateral Ligament:-

Ulna Collateral Ligament.

→ Consist of 2 parts

- 1) Anterior
- 2) Posterior

→ It is referred as anterior and posterior oblique

ligament and anterior and posterior bundles.

→ bundles are separated into anterior & posterior bands

→ They provides medial joint stability.

→ ~~Anterior~~ ^{Posterior} bundle extends from the posterior aspect of the medial epicondyle of the humerus to attach to the ulnar coronoid and olecranon process.

Functions -

→ Stabilizes against valgus torques at the medial elbow.

→ limits extension at the end of the elbow extension range of motion.

→ Guides joint motion throughout flexion range of motion.

→ Provides some resistance to longitudinal distraction of joint surface.

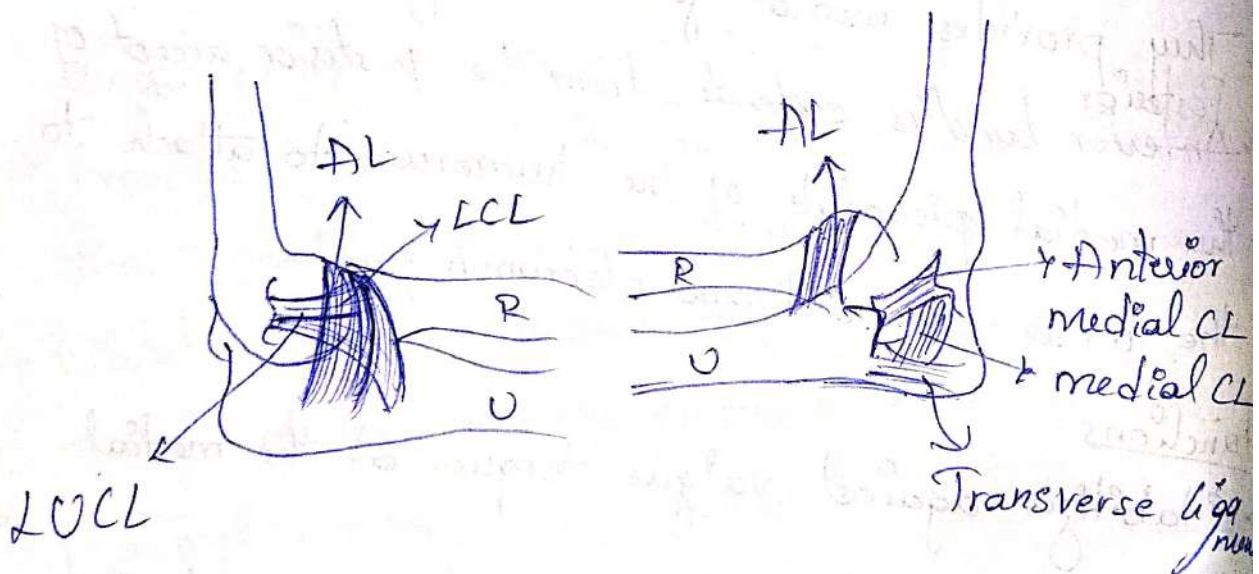
2) Lateral collateral Ligament (Radius) -

→ It includes lateral collateral ligament and (radial) lateral ulnar collateral ligament.

— Annular ligament.

→ Lateral radial collateral ligament is a fan-shaped structure that extends from the inferior aspect of the lateral epicondyle of the humerus to merge with the annular ligament.

→ Lateral Ulnar collateral ligament extended from the lateral humeral epicondyle to the lateral aspect of the ulna and the annular ligament. (Head of the radius)



Functions -

- Stabilizes elbow against varus torque.
- Stabilizes radial head,

Axis and plane

- Sagittal plane
- Coronal axis
- Axis is not fixed during movement so called as Screw displacement axis
- Instantaneous axis of rotation

Carrying angle

- The medial aspect of the trochlea extends more distally than the lateral aspect.
- This normal valgus angulation is called as carrying angle.
- Males - 11°
- Females - 15° (13°)
- ROM of elbow - 0° - 135° (flexion)

Osteokinematics

Flexion

- 1) Head of radius is gliding over capitulum anteriorly
- 2) Coronoid process of ulna gliding anteriorly.
- 3) Head of radius is making shelf over radial fossa
- 4) Coronoid fossa takes place in coronoid process.

Muscles - Brachio

- Biceps & radialis helps in stability..
- Brachialis helps in mobility.
- During 100° of flexion there is brachialis having max momentum.
- Brachialis is one joint muscle.
- Biceps is 2 joint muscle so position of shoulder effect the biceps.
- Brachioradialis will come into action when forearm is in mid-prone position.
- Momentum is 110°

Extension -

- The triceps is active concentrically to extend the elbow when the triceps acts in a closed kinematic chain.
- It acts as a synergist to prevent flexⁿ of the elbow when biceps is acting as a supinator.
- Other extensor of the elbow, the anconeus, assist in elbow extension & apparently acts as stabilizer during supination & pronation.

Proximal & distal radio-ulnar joints

① Proximal / superior radio-ulnar joint -

→ Uniaxial synovial variety of pivot joint.

Articulating Surfaces -

→ Ulnar radial notch, the annular ligament, head of the radius and the humeral capitulum.

② Distal / inferior radio-ulnar joint -

Articulating Surfaces -

→ Ulnar notch of the radius, the articular disc, and head of the ulna.

→ The articular disc has 2 articulating surfaces :
the proximal (superior) surface and distal (inferior) surface.

→ The proximal surface articulates with ulnar head.

→ The distal surface articulates with carpal bones.

Ligaments

→ 3 Ligaments

1) Annular ligaments -

→ It is a strong band that forms $4/5^{th}$ of a ring that encircles the radial head.

→ The proximal border of annular ligament blends with the joint capsule.

2) Quadrato Ligament -

- Extends from the inferior edge of ulna's radial notch to insert in the neck of the radius.
- It limits the spin of the radial head in supination and pronation.

3) Oblique Cord -

- 3) Oblique Cord -
- It is a flat band on the ventral forearm.
 - Extends from an attachment just inferior to the radial notch on the ulna to insert just below the bicipital tuberosity on the radius.

Osteokinematics -

- Supinations & pronation
- Axis of motion : Longitudinal axis
- ROM : Supinatⁿ - 0° - 70°
pronatⁿ - 0° - 90°
- 1° of freedom of motion is available.

Muscles -

- The primary muscles associated with the radioulnar joints are the pronator ~~triceps~~ } pronators.
pronator quadratus }
Biceps brachii }
Supinator } supinators.

Stability -

- Muscles are static stability.
- Pronator quadratus is major muscle.
- Provides dynamic stability & maintain stability throughout the movement.

Mobility -

- Triangular disc at inferior joint helps in supination & pronation.
- The complete separation of ulna when the forearm is in a neutral position is a diarthrodial joint.
- Functional activities like braking etc are used by wrist & forearm movements.
- To achieve any simple task 100° of forearm rotation.

Injuries -

- Dislocation of radial head (in children) & radial head from annular ligament.
- Distraction: injury to ligaments & muscles.
- Vagus & valgus.
- Tear (due to inflammation).